

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

INTRODUCTION TO POLITICAL RESEARCH METHODS

Instructor: Dr. Timea Balogh
Office Hours: Thursdays, 10am-12pm (and by appointment)
Class Information: Mondays & Wednesdays, 3-4:15 pm, in Wellman Hall 123

COURSE DESCRIPTION

How do we do the “science” part of political science? The purpose of this course is to help you start answering this question and to build your skills for conducting, interpreting, and presenting political science research. In this course, you will learn foundational research and data collection methods, how to create and evaluate research designs, and both quantitative and qualitative techniques for measuring political science concepts, hypothesis testing, and empirical analysis. Some of these terms may be confusing right now, and that’s okay, but by the end of this course, you will feel much more confident in your understanding. We will explore these skills in the context of a range of political science topics such as representation, democracy, and conflict, with plenty of opportunities for you to collaboratively identify, construct, and study research questions that you're passionate about. While the course will include learning and using the R programming language, no previous statistical or programming experience is needed. The only expectation is that you are curious, engaged, and prepared to gain the skills needed to confidently approach political science research.

IMPORTANT CLASS INFORMATION

Contacting Me

Throughout the quarter, you will likely encounter situations where you need to contact me via email or by attending my office hours. The following outlines what each mode of communication entails and a general recommendation for which activities are best suited to each.

Office Hours

Office hours are when I set aside two hours each week to be available to students. Office hours are often underutilized by students in college, even though I find they strongly correlate with better performance and engagement in the course. Consider this your time. You do **not** need an appointment to come by but are welcome to make one by emailing me.

My office hours are indicated at the top of this syllabus. If you prefer to meet outside of these hours and/or in-person, I am happy to schedule an alternate meeting with you, just email me.

Office hours are best for...

- Discussing or clarifying concepts or lecture material in greater detail
- Reviewing readings and their key arguments/concepts
- Brainstorming paper ideas or going over drafts and outlines
- Reviewing assignment grades
- Conversing about current events or comparative politics

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

- Providing constructive feedback on the course

I strongly encourage everyone to review the following document on the **hidden rules for office hours** - <https://www.wm.edu/sites/neurodiversity/documents/hidden-rules-for-office%20hours.pdf>

Email

If your question cannot wait until office hours or class, then you can use email to contact me. When emailing, I encourage you to introduce yourself and try to be as clear as possible with your inquiry so that I can best help you. To ensure that I see your email, please include the **class name** in your email subject line.

Email is best for...

- Asking quick, clarifying questions about lecture material or class assignments
- Setting up a meeting with me outside of office hours

Email is not a good alternative for having a direct conversation or an in-depth discussion, or exhaustively reviewing material from lecture. That is better suited for office hours.

I will do my best to respond to all emails within 24 hours. I usually reply to emails during standard “business hours” (9-5 Monday-Friday), so if you send an email very late or on a weekend, responses may take longer. If I have not responded within a reasonable time frame, please do not hesitate to send a follow-up email; I appreciate the reminder.

Canvas & Accessing Course Material

All readings and necessary documents for this course can be accessed through the Canvas page for the course. This is also where you will submit many of your assignments. I will often send messages and make announcements using Canvas, so please make sure your information is up to date and that you have notifications turned on or are otherwise regularly checking Canvas.

You are welcome to use Canvas messaging to contact me instead of email, although if you are trying to contact multiple people at the same time (other students, multiple professors, etc...) I ask that you use email instead to make “replying all” easier.

Required Textbook

There are two required texts you will need in this course, both of which you can purchase at the UC Davis Bookstore or another retailer, such as Amazon.

1. “Mastering ‘Metrics: The Path from Cause to Effect” by Joshua D. Angrist and Jörn-Steffen Pischke.
2. “Data Analysis for Social Science: A Friendly and Practical Introduction” by Elena Llaudet and Kosuke Imai

All other supplementary texts or articles will be free and open-access, or I will make selected chapters available as needed for the course. You do not need to purchase copies of any of these supplementary texts unless you would like to do so.

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

Required Software (R) & How to Access It

In this class we will be using a *free*, open-source programming language called R. R is geared towards statistical computing, data manipulation, and data visualization and is widely used by academics, statisticians, and data scientists for a range of tasks, from basic data manipulation to complex statistical modeling. Beyond this course, having R accessible on your computer and knowing how to use it will prove useful in your academic career and hopefully beyond, allowing you to conduct your own scientific research. We will be using RStudio, a coding environment, to access R.

There are two ways to get the latest version of R up and running:

1. Quick and easy (preferred) way: Run RStudio in your browser
2. Harder (alternative) way: Download and install R & RStudio on your machine

Option #1: RStudio in your browser (preferred)

This option does not require you to install anything and is completely free.

1. Go to <https://posit.cloud/>
2. Click on “Sign up” and create a free account
3. Click on “New Project” and select “New Project from Template”
4. Select “Data Analysis in R with the Tidyverse” and click “OK”
5. Voila, you’re all done!

Option 2: RStudio on your computer

This option is also completely free, but a bit more involved.

First, install R (the engine).

1. If you have Windows, click here to download the most recent version of R for Windows.
2. If you have Mac, click here to download the most recent version of R for Mac.
3. If you have Linux, click here to download the most recent version of R for Linux.
4. Check your downloads folder, double click the downloaded file, and install the file like you would any other program.

Then, install RStudio (think of this as your R dashboard). To do this, follow the steps below.

Once RStudio is installed, you will not need to open or use R, you will only use RStudio.

RStudio automatically uses R if it is installed on your computer.

1. Go to: <https://posit.co/download/rstudio-desktop/>
2. Scroll down and download the most recent version for either Mac or Windows
3. Check your downloads folder

Once you have both installed, double click on RStudio to run it and you’re good to go!

Assignments and Grading

Your grade for this course will be calculated as follows:

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

Class Participation (20%): Based on class attendance, (mostly) bi-weekly Canvas quizzes due before class on Wednesdays, as well as other tasks.

Problem Sets (30%): Three problem sets (10% each), due on Sundays by 11:59 pm, on Canvas.

Final Problem Set (25%): A slightly longer, more comprehensive problem set due during finals week, in lieu of a final exam.

Midterm Exam (25%): Takes place during finals week.

Extra Credit: A total of 3 points of extra credit will be possible for completing the class surveys (1 point per survey)

MORE DETAILS ON CLASS INFORMATION

Course Assignments

The assignments for this course consist of class participation, three standard problem sets, a midterm exam, and a slightly longer final problem set. There are also three surveys throughout the quarter for which you will earn extra credit for completing.

- **Class Participation (20%):** This part of your grade consists of a couple of different components. First, you should attend all lectures that you are able, as well as actively participate in any class activities, as I have found this is the best way to engage in your own learning. If you must miss a lecture, coming to office hours or otherwise scheduling a time to meet with me to discuss the course material will count positively towards this part of your grade. Second, there will be four (mostly) bi-weekly Canvas quizzes that will be due by the start of each class on Wednesday and contain material from both lecture and readings.
- **Problem Sets (total 30%, 10% each):** The best way to learn is by doing, which involves consistently practicing what you have learned in class and read about in the readings. This is especially true for learning new methodological tools and a programming language such as R. Therefore, there will be three standard problem sets throughout the course, which will focus on the current class content. They will be posted after class on Monday and will be due by the end of the week (by Sunday at 11:59 pm). All problem sets should be uploaded to Canvas by the due date in “pdf” form. They are graded out of 3 points, and late submissions will be penalized by 1 point for every 24 hours they are past due. If you anticipate needing to request an extension, you must do so when the problem set is posted, or ideally sooner.
- **Final Problem Set (25%):** Instead of a final exam, you will have a slightly longer and more comprehensive problem set due during finals week, by midnight on the date of the schedule final exam. The format and expectations will mirror those of the previous problem sets as described above. There will be no extensions available for this final

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

assignment, but it will be posted in advance, and you should have plenty of time to work on it on your own time.

- **Midterm Exam (25%):** There is one exam, scheduled to take place during Week 5, at the end of Module 2. The exam will be cumulative (covering Modules 1 & 2) and open note. This means you are allowed to bring any **print** material that you yourself have compiled, including any printouts of lecture slides or readings. Nothing electronic will be permitted, so if you take notes digitally (on a laptop or an iPad) you must print out these notes in advance. You can bring as little or as much material as you would like, but remember, to do well you will need to be able to both quickly access and readily understand the material to apply it to the exam questions. To this end, I highly recommend you organize your materials and make sure you thoroughly study the contents of papers and class content prior to the exam date.
- **Class Surveys (extra credit, up to 3 points):** I will send out three surveys throughout the quarter to gauge your progress and feelings about the course. This exercise is designed to make sure the course is meeting its objectives, your needs, and to allow you the opportunity to reflect on your own learning experience. This should not be thought of as an assessment or examination and will not count against you. The 2nd and 3rd surveys will follow a “start, stop, continue” format, where you will be asked to express what you think we as a class should **start** doing, **stop** doing, and **continue** doing going forward. Remember, the “class” does not only include my role as the instructor, but also and importantly includes your active role as a student and member of the classroom, so I expect that you will provide feedback on both aspects. To thank you for your time, you will receive 1 point of extra credit for completing each survey.

Inclusive Learning Statement

Your success in this class is important to me. It is my responsibility to ensure that you have the resources you need to be able to succeed in this course. Some of us may require accommodations because we all learn differently. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible so that we can develop a plan to meet both your needs and the requirements of the course and so that I can best help you. You do not need to reveal any details that you do not want to and know that I will never judge you, but I cannot read your mind or infer what is going on in your life. There is always something that can be done to help you and to make sure you are able to successfully complete the course, so please come talk to me as early as possible.

If you need official accommodations, you have a right to have these met. I encourage you to visit the UC Davis Student Disability Center website at sdc.ucdavis.edu for more information regarding the resources available to you and to determine how you could improve your learning as well. This syllabus and other course materials can be made available in alternate formats.

There are also a range of resources on campus, including the Writing Center and the Academic Assistance and Tutoring Centers, which assist students by helping them improve their study skills, academic writing, and other specific topics.

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

Course Outline

Below you will find the schedule for when we will be covering which topics, as well as the deadlines for readings and other assignments. All readings are due on the day that they are listed, so please read each day's readings before class on the specified day. On the other hand, other assignments are listed as "Do" and are not necessarily listed on the day that they are due. Instead, their place on the syllabus should serve as a reminder that a deadline is coming up soon, with the actual due date provided in parentheses. You can find this information, with associated links to readings and assignments, on the course Canvas page. If any changes are made to the below schedule, I will let you know sufficiently in advance and update this schedule on Canvas.

Module 1: Questions, Concepts, and Measures

Week 1, Monday: Introduction to the course

Read: This syllabus

Week 1, Wednesday: Introduction to the (social) scientific process

Read: Kellstedt and Whitten, Chapter 1: The Scientific Study of Politics

Do: Start of Quarter Introductory Survey (due by Friday, 11:59 pm, on Canvas)

Week 2, Monday: Asking good research questions

Read: Alvesson, M., & Sandberg, J. (2011). Generating research questions through problematization. *The Academy of Management Review*, 36(2), 247–271.

Read: Barakso, Maryann, Daniel M. Sabet, and Brian F. Schaffner (2014): "The Research Question."

Do: Quiz 1 (due by Wednesday, before class, on Canvas)

Week 2, Wednesday: What's in a theory? Correlation vs. Causation

Read: Kellstedt and Whitten, Ch. 2: The Art of Theory Building

Read: Barakso, Maryann, Daniel M. Sabet, and Brian F. Schaffner (2014): "Linking Theory and Inference."

Do: Problem Set 1 (due by Sunday, 11:59 pm, on Canvas)

Week 3, Monday: Measuring concepts

Read: Kellstedt and Whitten, Ch. 5: Measuring Concepts of Interest

Module 2: Types of (Quantitative) Research Designs

Week 3, Wednesday: Experiments and Quasi-Experiments

Read: "Experimental and Quasi-Experimental Designs for Research" by Donald T. Campbell and Julian C. Stanley, p. 6-26

Read: Angrist and Pischke, p. 11-24

Week 4, Monday: Natural Experiments

Listen to this podcast on The Oregon Health Insurance Experiment:

<https://freakonomics.com/podcast/does-health-insurance-make-you-healthier/>

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

Do: Quiz 2 (due by Wednesday, before class, on Canvas)

Week 4, Wednesday: Regression Discontinuity

Read: Angrist and Pischke, Ch. 4.1: Birthdays and Funerals

Do: Problem Set 2 (due by Sunday, 11:59pm, on Canvas)

Week 5, Monday: Differences-in-Differences (DiD)

Read: Angrist and Pischke, Ch. 5.2: Drink, Drank, ...

Week 5, Wednesday: In-Class Midterm Exam

Read: Nothing

Do: Get R up and running on your device (due by next class; see syllabus section above for instructions on how to do this)

Module 3: Data Analysis

Week 6, Monday: Getting Started with R

Read: Llaudet and Imai, Ch. 1.5 and 1.6 & Kellstedt and Whitten, Ch. 3

Do: Quiz 3 (due by Wednesday, before class, on Canvas)

Week 6, Wednesday: Finding and Cleaning Data

Explore: <https://guides.nyu.edu/polisci/political-data-and-statistics>,
<https://guides.nyu.edu/polisci/public-opinion-data>, <https://guides.nyu.edu/polisci/electoral-data>,
and <https://guides.nyu.edu/polisci/political-conflict-data>

Then read: Kellstedt and Whitten, Ch. 6: Getting to Know Your Data

Do: Mid-Quarter Check In Survey (due by Friday, 11:59 pm, on Canvas)

Week 7, Monday: Describing Variables

Read: Llaudet and Imai, Ch. 1.7 and 1.8

Week 7, Wednesday: Statistical Analysis I

Read: Angrist and Pischke, Chapter 2, “Regression”

Week 8, Monday: Statistical Analysis II

Read: Llaudet and Imai, Ch. 4.1

Do: Quiz 4 (due by Wednesday, before class, on Canvas)

Week 8, Wednesday: Data Visualization

Watch: David McCandless, TED

Talk, https://www.ted.com/talks/david_mccandless_the_beauty_of_data_visualization?language=en#t-106992

Do: Problem Set 3 (due by Sunday, 11:59 pm, on Canvas)

Module 4: Qualitative analysis IS empirical analysis

Sample Syllabus – Intro to Research Methods

Timea Balogh, Dept. of Government and Public Policy, University of Strathclyde

Week 9, Monday: What is empirical?

Read: Tilda Gaskell, "The Process of Empirical Research: a learning experience?"

Read: Gerring, John. 2012. "Mere Description." *British Journal of Political Science* 42(4): 721–46. doi: 10.1017/S0007123412000130.

Week 9, Wednesday: The Case Study Method

Read: Geddes, Barbara. "How the Cases You Choose Affect the Answers You Get: Selection Bias in Comparative Politics." *Political Analysis* 2:1 (1990), pp. 131-50.

Week 10, Monday: Interviews & Ethnography

Read: Nichols, Michael. *The Lost Art of Listening*. New York: The Guilford Press (1995), pp. 61-68, 125-35.

Read: Fenno, Richard. *Home Style: House Members in Their Districts*. Boston: Little, Brown, and Company (1978).

Week 10, Wednesday: Research Ethics

Read: Pacheco-Vega, R., & Parizeau, K. (2018). Doubly Engaged Ethnography: Opportunities and Challenges When Working With Vulnerable Communities. *International Journal of Qualitative Methods*, 17(1).

Read: Malesky, Edmund, Paul Schuler, and Anh Tran. 2012. "The Adverse Effects of Sunshine: A Field Experiment on Legislative Transparency in an Authoritarian Assembly." *American Political Science Review* 106(4): 762–86.

Optional: Barsbai, Toman and Bartos, Vojtech and Licuanan, Victoria and Steinmayr, Andreas and Tionson, Erwin R. and Yang, Dean and Yang, Dean, *Picture this: Social Distance and the Mistreatment of Migrant Workers* (2022). CESifo Working Paper No. 10177.

Do: End of Quarter Exit Survey (due by Friday, 11:59 pm, on Canvas)

Do: Final Problem Set (due by next week Thursday, 11:59 pm, on Canvas)

Finals Week, Thursday: Final Problem Set Due, by 11:59 pm, on Canvas